

Quantum Information Theory

Uhlmann's Theorem

(with a Summary of the Relevant Properties of Fidelity)

Talk 4 — July 16, 2026

Part I — Summary of the Relevant Properties of Fidelity

We use the squared-fidelity convention

$$F(\rho, \sigma) = \|\sqrt{\rho} \sqrt{\sigma}\|_1^2, \quad \text{with root fidelity} \quad \sqrt{F(\rho, \sigma)} = \|\sqrt{\rho} \sqrt{\sigma}\|_1,$$

versus the trace distance

$$T(\rho, \sigma) = \frac{1}{2} \|\rho - \sigma\|_1.$$

Fidelity is a normalized measure of similarity, with larger values representing greater overlap.

1. Boundedness.

$$0 \leq F(\rho, \sigma) \leq 1.$$

2. Equality condition.

$$F(\rho, \sigma) = 1 \iff \rho = \sigma.$$

Two normalized quantum states have maximum fidelity exactly when they are the same state.

3. Orthogonality condition.

$$F(\rho, \sigma) = 0 \iff \text{supp}(\rho) \perp \text{supp}(\sigma).$$

Fidelity vanishes exactly when the states have orthogonal supports and are therefore perfectly distinguishable. Here $\text{supp}(\rho) = (\ker \rho)^\perp$ is the linear subspace on which ρ maps to nonzero vectors; its dimension is the rank of ρ .

4. Symmetry.

$$F(\rho, \sigma) = F(\sigma, \rho).$$

Fidelity does not depend on the order in which the two states are compared.

5. Pure-state formula.

$$F(|\psi\rangle, |\phi\rangle) = |\langle\psi|\phi\rangle|^2.$$

Fidelity reduces to the ordinary squared overlap (transition probability) between the two state vectors.

6. Pure–mixed formula. If one state is pure, then

$$F(|\psi\rangle, \rho) = \langle \psi | \rho | \psi \rangle.$$

This is the probability that ρ passes the projective test for being the target state $|\psi\rangle$.

7. Commuting (classical) case. If ρ and σ share an eigenbasis,

$$\rho = \sum_x p_x |x\rangle\langle x|, \quad \sigma = \sum_x q_x |x\rangle\langle x|,$$

then

$$F(\rho, \sigma) = \left(\sum_x \sqrt{p_x q_x} \right)^2.$$

Quantum fidelity reduces to the square of the classical Bhattacharyya overlap when the states commute.

8. Uhlmann’s theorem.

$$F(\rho, \sigma) = \max_{\substack{|\psi_\rho\rangle, |\psi_\sigma\rangle \\ \text{purifications of } \rho, \sigma}} |\langle \psi_\rho | \psi_\sigma \rangle|^2.$$

Fidelity is the largest possible squared overlap between purifications of the two mixed states. *(Proved in Part II.)*

9. Unitary and isometric invariance. For every unitary or isometry V ,

$$F(V\rho V^\dagger, V\sigma V^\dagger) = F(\rho, \sigma).$$

Reversible embeddings and unitary evolution preserve fidelity exactly.

10. Multiplicativity under tensor products.

$$F(\rho_1 \otimes \rho_2, \sigma_1 \otimes \sigma_2) = F(\rho_1, \sigma_1) F(\rho_2, \sigma_2).$$

The fidelity of product states is the product of the fidelities of their individual components.

11. Monotonicity under quantum channels. For every quantum channel $\mathcal{N}$,

$$F(\mathcal{N}(\rho), \mathcal{N}(\sigma)) \geq F(\rho, \sigma).$$

Applying the same physical process to both states cannot decrease their fidelity; noise can make states more similar.

12. Monotonicity under partial trace. For bipartite states, with $\rho_A = \text{Tr}_B \rho_{AB}$ and $\sigma_A = \text{Tr}_B \sigma_{AB}$,

$$F(\rho_A, \sigma_A) \geq F(\rho_{AB}, \sigma_{AB}).$$

Discarding a subsystem cannot decrease fidelity, because it removes potentially distinguishing information.

- 13. Behaviour under measurements.** For a POVM $\{\Lambda_k\}_k$ define $p_k = \text{Tr}(\Lambda_k \rho)$ and $q_k = \text{Tr}(\Lambda_k \sigma)$. Then

$$\sum_k \sqrt{p_k q_k} \geq \sqrt{F(\rho, \sigma)}.$$

Every measurement produces classical distributions whose overlap is at least the root fidelity of the original quantum states.

- 14. Fuchs–van de Graaf inequalities.** With trace distance $T(\rho, \sigma) = \frac{1}{2} \|\rho - \sigma\|_1$,

$$1 - \sqrt{F(\rho, \sigma)} \leq T(\rho, \sigma) \leq \sqrt{1 - F(\rho, \sigma)}.$$

- 15. Exact pure-state relation with trace distance.** For pure states,

$$T(|\psi\rangle, |\phi\rangle) = \sqrt{1 - F(|\psi\rangle, |\phi\rangle)}.$$

Fidelity and trace distance define compatible notions of closeness and allow estimates in one measure to be converted into estimates in the other; for pure states they determine one another exactly.

- 16. Purified distance.**

$$P(\rho, \sigma) = \sqrt{1 - F(\rho, \sigma)}.$$

Purified distance converts fidelity into a genuine distance that behaves especially well with purifications and approximate quantum states.

- 17. Contraction of purified distance under channels.**

$$P(\mathcal{N}(\rho), \mathcal{N}(\sigma)) \leq P(\rho, \sigma).$$

Physical processing cannot increase purified distance, just as it cannot increase trace distance.

- 18. Fidelity itself is not a metric.** Fidelity is a similarity measure rather than a distance: identical states have fidelity 1, and fidelity itself does not satisfy the usual metric axioms such as the triangle inequality.

Part II — Uhlmann's Theorem

We have the definition of (squared) fidelity between two states on A ,

$$F(\rho, \sigma) = \|\sqrt{\rho}\sqrt{\sigma}\|_1^2,$$

and we want to prove that, equivalently,

$$F(\rho_A, \sigma_A) = \max_{\text{purifications}} |\langle \psi_\rho | \psi_\sigma \rangle|^2,$$

where the maximum is over all purifications $|\psi_\rho\rangle_{RA}$ of ρ_A and $|\psi_\sigma\rangle_{RA}$ of σ_A , with an extended reference system $R \otimes A$ and $\dim R = \dim A$.

Side definitions and identities first:

- 1) Definition of a maximally entangled state, $|\Gamma\rangle$.
- 2) Identity: $\langle \Gamma | M_R \otimes N_A | \Gamma \rangle = \text{Tr}(M^T N)$.
- 3) Using $|\Gamma\rangle$ to purify states.
- 4) Alternative trace 1-norm definition: $\|X\|_1 = \max_{V: V^\dagger V = I} |\text{Tr} V X|$.

1 Maximally entangled state

A maximally entangled state:

$$|\Gamma\rangle_{RA} = \sum_{j=1}^d |j\rangle_R |j\rangle_A \quad \text{on } R \otimes A,$$

where $\{|j\rangle_R\}_{j=1}^d$ and $\{|j\rangle_A\}_{j=1}^d$ are arbitrary orthonormal bases for R and A respectively. We use the *unnormalized* version above as an algebraic tool; properly we should divide by $\sqrt{d}$.

Why “maximally entangled”? Take the normalized version

$$|\Gamma'\rangle_{RA} = \frac{1}{\sqrt{d}} \sum_j |j\rangle_R |j\rangle_A, \quad \left(\text{compare the } d=2 \text{ Bell state } |\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \right).$$

Then

$$\begin{aligned} |\Gamma'\rangle_{RA} \langle \Gamma'|_{RA} &= \frac{1}{d} \sum_{ij} |j\rangle |j\rangle \langle i| \langle i| = \frac{1}{d} \sum_{ij} (|j\rangle_R \otimes |j\rangle_A) (\langle i|_R \otimes \langle i|_A) \\ &= \frac{1}{d} \sum_{ij} |j\rangle \langle i|_R \otimes |j\rangle \langle i|_A. \end{aligned}$$

Now trace out system R via $\text{Tr}_R = \text{Tr} \otimes I_A$ (trace on R , leave A alone):

$$(\text{Tr} \otimes I) |\Gamma'\rangle \langle \Gamma'| = \frac{1}{d} \sum_{ij} \underbrace{\text{Tr} |j\rangle \langle i|_R}_{\delta_{ij}} \otimes |j\rangle \langle i|_A = \frac{1}{d} \sum_i |i\rangle \langle i|_A = \frac{1}{d} I_A,$$

and similarly $\text{Tr}_A |\Gamma'\rangle \langle \Gamma'| = \frac{1}{d} I_R$. So each subsystem in isolation is maximally (completely) mixed as a single-particle state. This is the definition of maximal entanglement between two systems.

2 The identity $\langle \Gamma | M_R \otimes N_A | \Gamma \rangle = \text{Tr}(M^T N)$

(Note: M^T is the transpose, *not* the adjoint $M^\dagger$.)

2.1 Transpose trick for $|\Gamma\rangle$

$$\boxed{(M \otimes I) |\Gamma\rangle = (I \otimes M^T) |\Gamma\rangle.}$$

Recall $A|j\rangle = \sum_i A_{ij} |i\rangle$; check: $\langle k | A | j \rangle = \sum_i A_{ij} \langle k | i \rangle = \sum_i A_{ij} \delta_{ki} = A_{kj}$. ✓

Now compare both sides:

$$\begin{aligned} \text{LHS} &= (M \otimes I) \sum_j |j\rangle \otimes |j\rangle = \sum_j M |j\rangle \otimes |j\rangle = \sum_{ji} M_{ij} |i\rangle \otimes |j\rangle, \\ \text{RHS} &= (I \otimes M^T) \sum_j |j\rangle \otimes |j\rangle = \sum_j |j\rangle \otimes M^T |j\rangle = \sum_j |j\rangle \otimes \sum_i [M^T]_{ij} |i\rangle \\ &= \sum_j |j\rangle \otimes M_{ji} |i\rangle = \sum_{ji} M_{ji} |j\rangle \otimes |i\rangle \quad (\text{swap } i \leftrightarrow j). \end{aligned}$$

Therefore LHS = RHS.

2.2 Proving the identity

We want $\langle \Gamma | (M \otimes N) | \Gamma \rangle = \text{Tr}(M^T N)$. Using the transpose trick,

$$\langle \Gamma | M \otimes N | \Gamma \rangle = \langle \Gamma | (M \otimes I) (I \otimes N) | \Gamma \rangle = \langle \Gamma | (I \otimes M^T) (I \otimes N) | \Gamma \rangle = \langle \Gamma | (I \otimes M^T N) | \Gamma \rangle. \quad (*)$$

But in general, for any operator X on A ,

$$\langle \Gamma | I \otimes X | \Gamma \rangle = \sum_i \sum_j (\langle i | \otimes \langle i |) (I \otimes X) (|j\rangle \otimes |j\rangle) = \sum_{ij} \underbrace{\langle i | j \rangle}_{\delta_{ij}} \langle i | X | j \rangle = \sum_i \langle i | X | i \rangle = \sum_i X_{ii},$$

so that

$$\boxed{\langle \Gamma | I \otimes X | \Gamma \rangle = \text{Tr } X.}$$

Applying this to (*) with $X = M^T N$,

$$\boxed{\langle \Gamma | M \otimes N | \Gamma \rangle = \text{Tr}(M^T N).} \quad \checkmark$$

3 Using $|\Gamma\rangle$ to purify states

Recall the previous method to purify a ρ_A :

$$\rho_A \longrightarrow |\psi_\rho\rangle_{RA} = \sum_i \sqrt{p_i} |i\rangle_R \otimes |e_i\rangle_A,$$

with $|i\rangle_R$ an arbitrary ONB of R , $|e_i\rangle_A$ the ONB of eigenvectors of ρ_A , and p_i its eigenvalues.

An *alternative* purification is

$$\boxed{|\psi_\rho\rangle_{RA} = (I_R \otimes \sqrt{\rho_A}) |\Gamma\rangle_{RA} .}$$

Check.

$$\begin{aligned}
|\psi_\rho\rangle_{RA}\langle\psi_\rho|_{RA} &= (I_R \otimes \sqrt{\rho_A}) |\Gamma\rangle\langle\Gamma| (I_R \otimes \sqrt{\rho_A}) \\
&= (I \otimes \sqrt{\rho}) \sum_{ij} |i\rangle\langle i| \langle j| \langle j| (I \otimes \sqrt{\rho}) \\
&= \sum_{ij} |i\rangle\langle j|_R \otimes \sqrt{\rho_A} |i\rangle\langle j|_A \sqrt{\rho_A}.
\end{aligned}$$

Trace out R :

$$\begin{aligned}
(\text{Tr} \otimes I_A) |\psi_\rho\rangle\langle\psi_\rho| &= \sum_{ij} \underbrace{\text{Tr} |i\rangle\langle j|}_{\delta_{ij}} \otimes \sqrt{\rho_A} |i\rangle\langle j|_A \sqrt{\rho_A} \\
&= \sum_i \sqrt{\rho_A} |i\rangle\langle i| \sqrt{\rho_A} = \sqrt{\rho_A} \left(\underbrace{\sum_i |i\rangle\langle i|}_{I_A} \right) \sqrt{\rho_A} = \rho_A. \quad \checkmark
\end{aligned}$$

4 Alternative trace 1-norm definition

$$\|X\|_1 = \max_{\substack{U \text{ unitary} \\ U^\dagger U = I}} |\text{Tr} UX|.$$

Think of this as a generalization of the scalar case for $\mathbb{C}$ -numbers. For $x \in \mathbb{C}$ and $u \in \mathbb{C}$ on the unit circle ($u = e^{i\theta}$), what is $\max_{u=e^{i\theta}} |ux|$?

$$|ux| = |u| \cdot |x| = |x| \quad (\text{since } |u| = 1 \ \forall u \in \{e^{i\theta}\}) \implies \max_{u=e^{i\theta}} |ux| = |x| \text{ trivially.}$$

One can show the operator statement for general operators on $\mathbb{C}^n$.

We now have the pieces necessary to prove Uhlmann's theorem.

5 Proof of Uhlmann's Theorem

Given the fidelity of two states ρ, σ on A defined as $F(\rho, \sigma) = \|\sqrt{\rho} \sqrt{\sigma}\|_1^2$, we show

$$F(\rho_A, \sigma_A) = \max_{\text{purifications}} |\langle\psi_{\rho_A}|\psi_{\sigma_A}\rangle|^2, \quad \rho_A \xrightarrow{\text{purif}} |\psi_{\rho_A}\rangle_{RA}, \quad \sigma_A \xrightarrow{\text{purif}} |\psi_{\sigma_A}\rangle_{RA}.$$

5.1 Step 1: choose the $|\Gamma\rangle$ purification

Choose (for now) arbitrary ONBs $\{|i\rangle_A\}$ for system A and $\{|i\rangle_R\}$ for reference system R , with $\dim R = \dim A$. Define the maximally entangled state

$$|\Gamma\rangle_{RA} = \sum_i |i\rangle_R |i\rangle_A,$$

and define the purifications

$$|\psi_\rho\rangle_{RA} = (I_R \otimes \sqrt{\rho}) |\Gamma\rangle_{RA}, \quad |\psi_\sigma\rangle_{RA} = (I_R \otimes \sqrt{\sigma}) |\Gamma\rangle_{RA}. \quad (*)$$

5.2 Step 2: reduce max over purifications to max over unitaries

Recall that all possible purifications of a given state are related by unitary transformations on the reference system. All possible purifications are given by

$$(U_R^1 \otimes I_A) |\psi_\rho\rangle_{RA}, \quad (U_R^2 \otimes I_A) |\psi_\sigma\rangle_{RA},$$

where U_R^1, U_R^2 are unitary on R ; all purifications are obtained by varying U_R^1, U_R^2 over $\mathcal{U}$ (all unitary operators). So all possible overlaps of purifications are

$$\langle \psi_\rho | (U_R^{1\dagger} \otimes I_A) (U_R^2 \otimes I_A) | \psi_\sigma \rangle = \langle \psi_\rho | (U_R^{1\dagger} U_R^2 \otimes I_A) | \psi_\sigma \rangle = \langle \psi_\rho | (U_R \otimes I_A) | \psi_\sigma \rangle,$$

where $U_R = U_R^{1\dagger} U_R^2$ is also unitary. Hence we only have to vary the single U_R to get all overlaps, and we can write

$$\boxed{\max_{\text{purifications}} |\langle \psi_\rho | \psi_\sigma \rangle| = \max_{U_R \in \mathcal{U}} |\langle \psi_\rho | (U_R \otimes I_A) | \psi_\sigma \rangle|}.$$

[Notes end here — the remaining steps combine the transpose-trick identity of §2 with the alternative trace-norm definition of §4 to evaluate this maximum as $\|\sqrt{\rho} \sqrt{\sigma}\|_1$, completing the proof.]